Machine Learning:

Logistic Regression

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Introduction

Classification

- **Make a decision:**
  - **Email: Spam / Not Spam?**
  - **Tumor: Malignant / Benign?**

- **Mathematical representation:**

  $y \in 0, 1$ **with**

  $y = 0$**: Negative class (e.g, benign tumor)**

  $y = 1$**: Positive class (e.g., malignant tumor)**

Odds and Odds Ratio

Probability

- Probability (p)
  - $p = P(Y = 1)$
  - It always lies between 0 and 1.
- Example
  - If $p = 0.8$, there is an 80% chance of success.

Odds

- Odds are defined as:

$$\text{odds} = \frac{p}{1-p}$$

- So odds compare "how likely somethings is to happen" vs. "how likely it is NOT to happen"

Example:

Probability p	Odds $\frac{p}{1-p}$	Interpretation
0.9	9	"9 to 1" in favor
0.8	4	"4 to 1" in favor
0.5	1	equally likely, yes vs no
0.2	0.25	"1 to 4" in favor

Odds

Key intuition:

- Odds $= 1$ means a $50/50$ event.
- Odds > 1 means event is more likely than not.
- Odds < 1 means event is less likely than not.

Odds Ratio (OR)

- Definition: An odds ratio compared the odds between two groups:

$$\text{odds ratio} = \frac{\text{odds in group A}}{\text{odds in group B}}$$

- So it answers the question:

  How many times larger are the odds in group A than in group B?

Odds Ratio

Example

- Suppose:
  - In the control group, $p = 0.2$.
  - In the treatment group, $p = 0.5$.
- Compute odds
  - Control group:
$$\text{odds} = \frac{0.2}{0.8} = 0.25$$
  - Treatment group:
$$\text{odds} = \frac{0.5}{0.5} = 1$$
- Compute odds ratio:
$$\text{odds ratio} = \frac{1}{0.25} = 4$$
- **Interpretation:** The treatment group has 4x higher odds of success than the control group.

Odds Ratio

Odds Ratio is NOT the same as probability ratio.

- Probability ratio (risk ratio)

$$\frac{p_1}{p_0}$$

- Odds ratio

$$\frac{p_1/(1-p_1)}{p_0/(1-p_0)}$$

- These are only similar when probabilities are small.

Odds Ratio: Example

Odds Ratio is NOT the same as probability ratio.

- Probability
  - Control: $p = 0.5$
  - Treatment: $p = 0.75$
- Risk ratio:
$$\frac{0.75}{0.5} = 1.5$$
- Odds
  - Control odds $= 0.5/0.5 = 1$
  - Treatment odds $= 0.75/0.25 = 3$
- Odds ratio
$$\frac{3}{1} = 3$$

There is a big difference between the risk ratio and odds ratio.

Logistic Regression Core Idea

Core Idea

Procedure

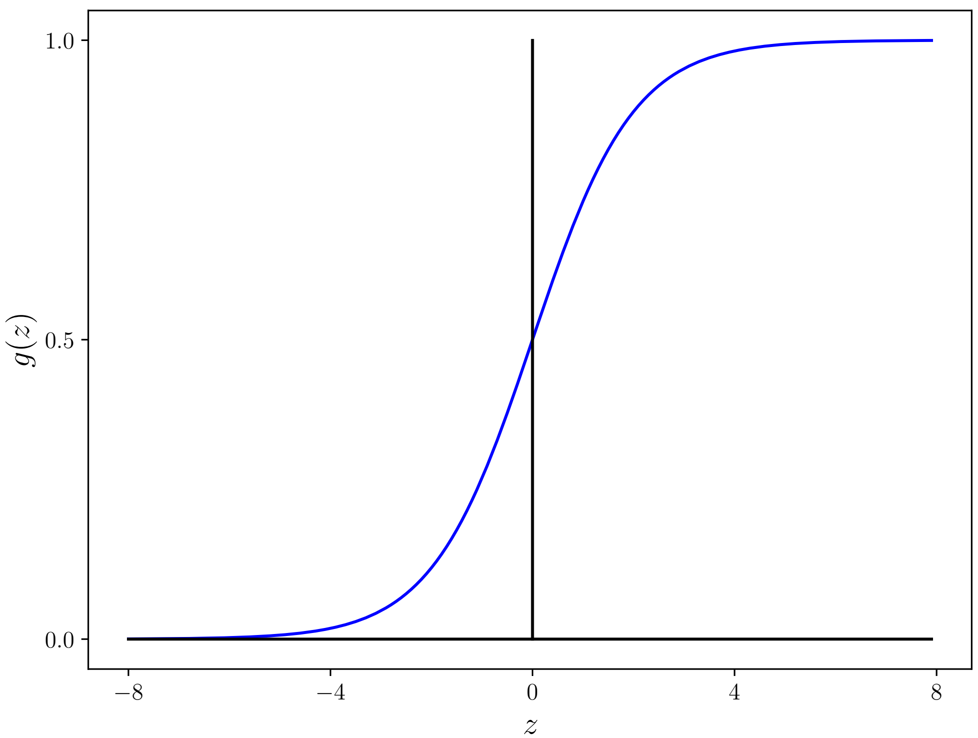
- Linearly combine the independent variables

$$z = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_d x_d$$

  where $z \in (-\infty, \infty)$.

- Signoid function to convert the real number z to $g(z) \in (0, 1)$:

$$g(z) = \frac{1}{1 + e^{-z}}$$



Core Idea

Procedure

- The full model:

$$P(y = 1|x) = p = g(z) = \frac{1}{1 + e^{-z}}$$

- This means:

$$z = \ln \frac{p}{1 - p}$$

Core Idea

- Odds

$$\text{odds} = \frac{p}{1-p}$$

- log-odds (logit)

$$\ln\left(\frac{p}{1-p}\right)$$

Logistic regression is linear in log-odds:

$$\ln\left(\frac{p}{1-p}\right) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_d x_d$$

Build Logistic Regression Models

Logistic regression model

- **We want**

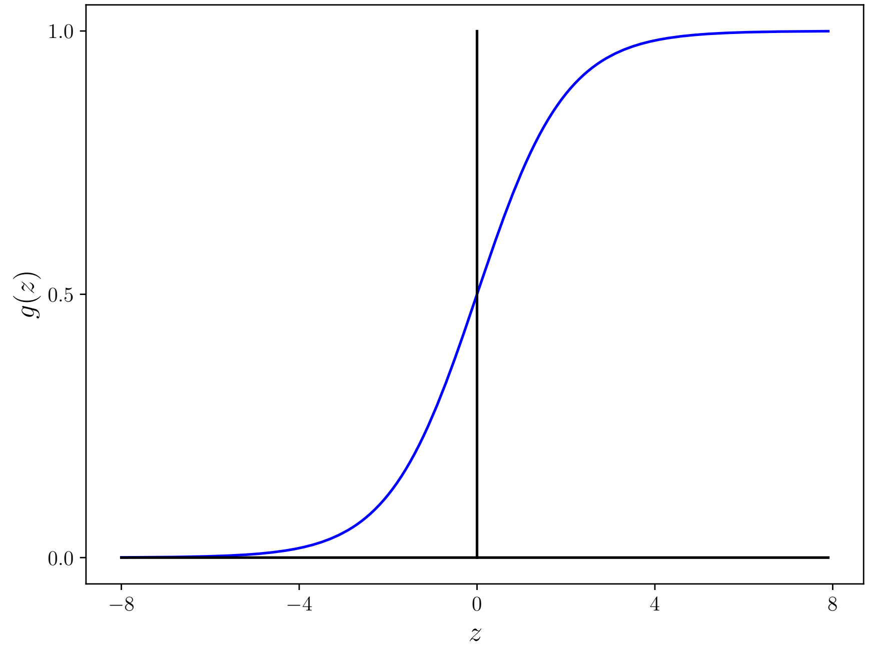
$$0 \le h_\theta(x) \le 1$$

- **This can be achieved by using the logistic/sigmoid function:**

$$h_\theta(x) = g(\theta_0 x_1 + \theta_1 x_1 + \cdots + \theta_n x_n) = g(\theta^T x)$$

with $g(z)$ **defined as**

$$g(z) = \frac{1}{1 + e^{-z}}$$



Interpretation of logistic regression output

- $h_\theta(x)$ **is interpreted as the estimated probability that** $y = 1$ **given** x**, i.e.**

$$\begin{aligned} h_\theta(x) &= g(\theta^T x) = P(y = 1|x;\theta) \\ 1 - h_\theta(x) &= g(\theta^T x) = P(y = 0|x;\theta) \end{aligned}$$

- **Example: If** $z = \theta_0 + \theta_1 x$ **where** x **is the tumor size and** $h_\theta(x) = 0.7$**, then we can tell the patient that there is a 70% probability that the tumor is malignant.**

- **Suppose we predict** $y = 1$ **when** $h_\theta(x) \geq 0.5$ **and predict** $y = 0$ **when** $h_\theta(x) < 0.5$**. This means that**

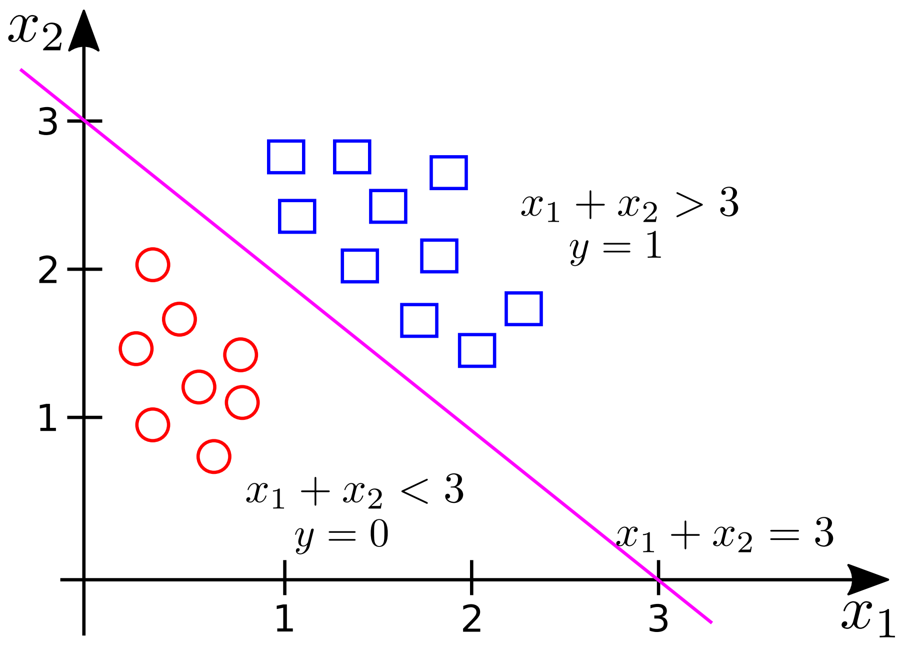
$$y = \begin{cases} 1 & \textbf{if } \theta^T x \geq 0 \\ 0 & \textbf{if } \theta^T x < 0 \end{cases}$$

Decision boundary

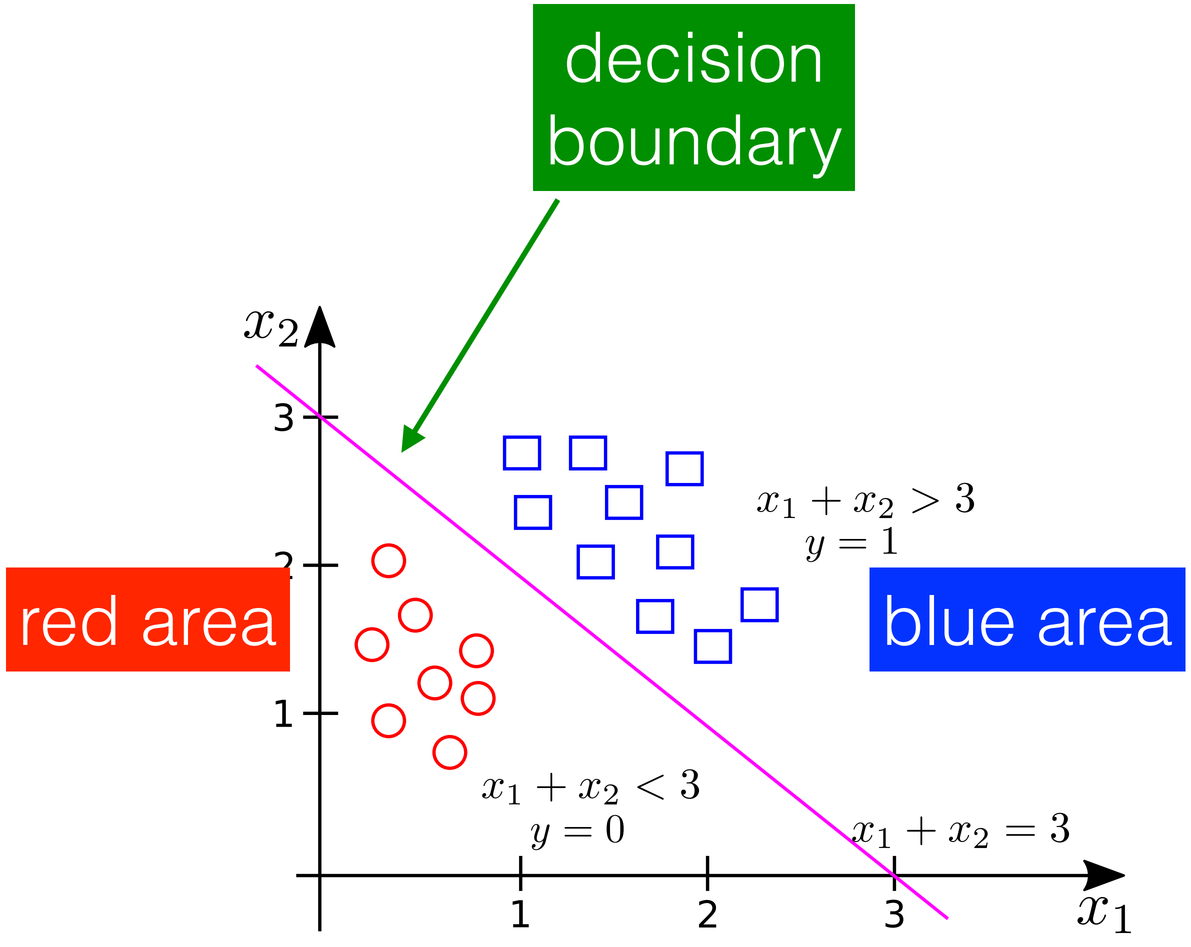
- $\theta^T x = 0$ **divides the feature space into two parts. The part where** $\theta^T x \geq 0$ **corresponds to** $y = 1$ **and the part where** $\theta^T x < 0$ **corresponds to** $y = 0$**.**

- **Example: Suppose** $h_\theta(x) = g(\theta_0 + \theta_1 x + \theta_2 x)$ **with** $\theta_0 = -3, \theta_1 = 1, \theta_2 = 1$**. Then**

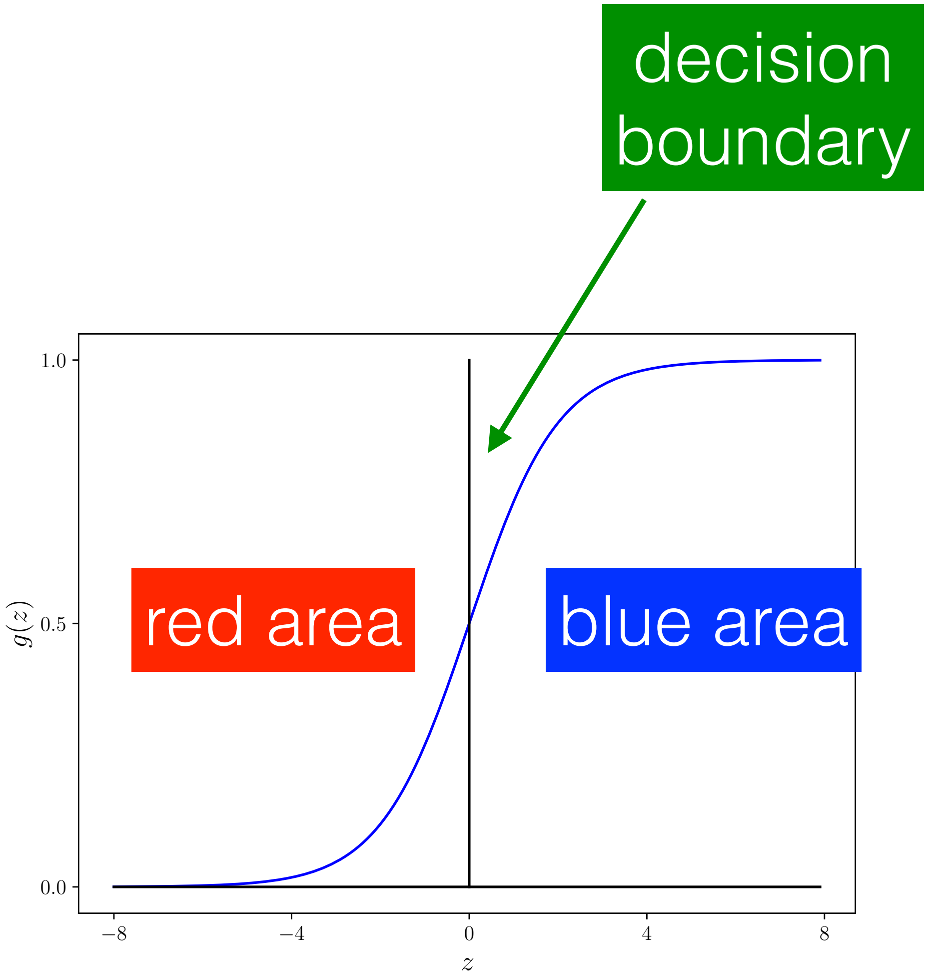
$\theta^T x = 0$	**corresponds to the magenta line**
$\theta^T x > 0$	**corresponds to the area above the magenta line**
$\theta^T x < 0$	**corresponds to the area below the magenta line**



Decision boundary



feature space



z space

Decision boundary

- As x changes, z changes linearly.
- Sigmoid maps z on the real line $(-\infty, +\infty)$ into probability $(0, 1)$.
- Decision boundary occurs when $z = 0$ and $p = 0.5$, i.e.

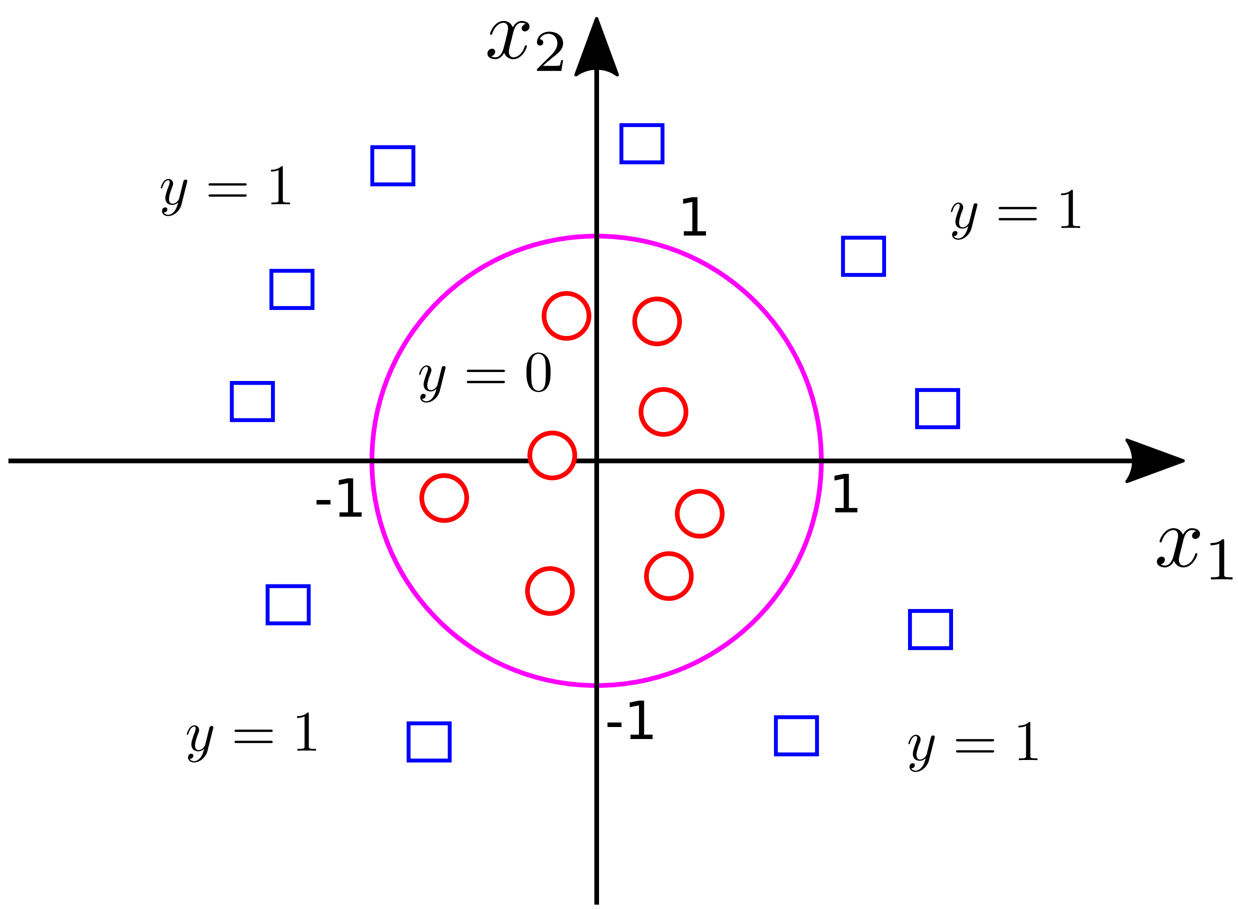
$$\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_d x_d = 0$$

- Log-odds convers probability (0, 1) to the real line $(-\infty, +\infty)$

$$\ln\left(\frac{p}{1-p}\right) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_d x_d$$

Nonlinear decision boundary

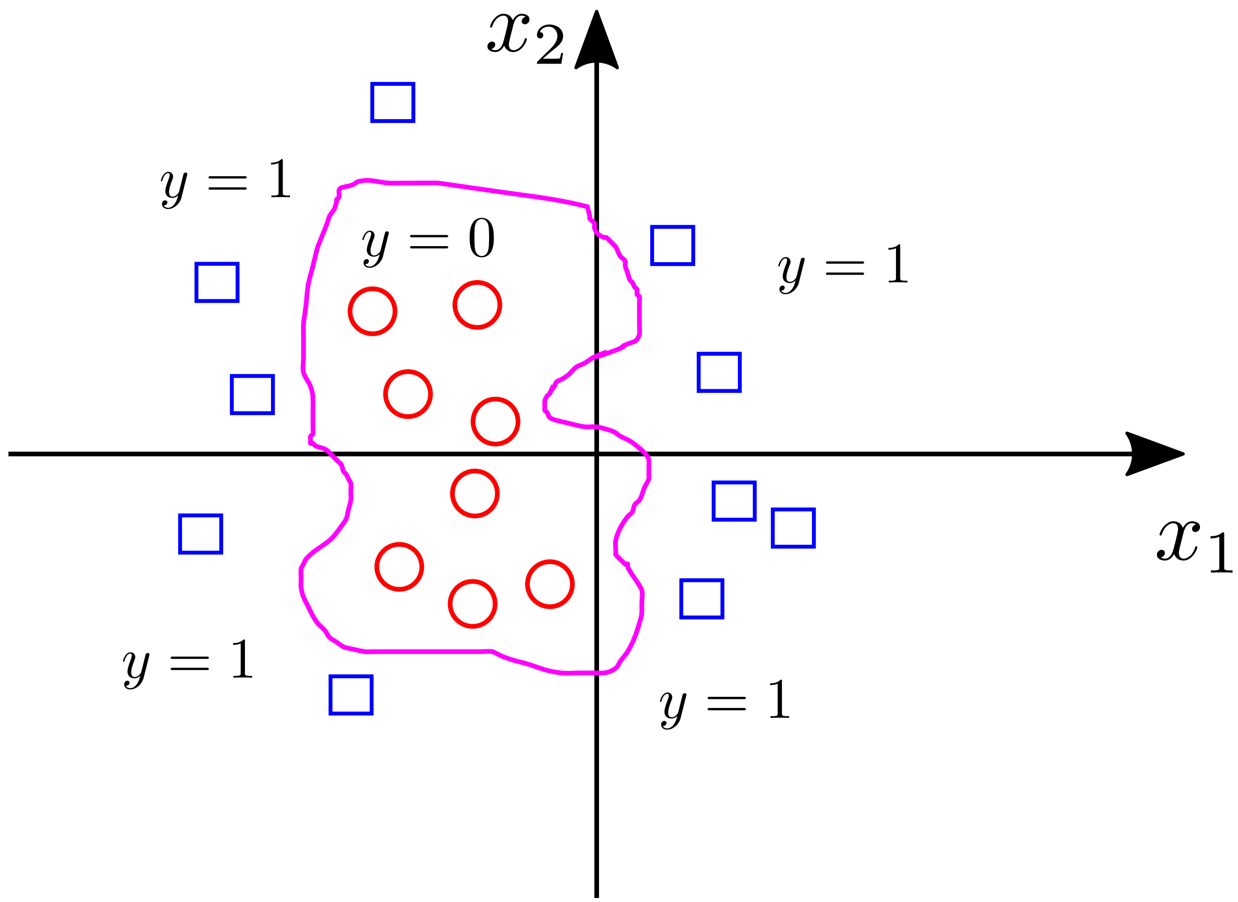
- **Example: Suppose** $h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3^2 + \theta_4 x_4^2)$ **with** $\theta_0 = -1, \theta_1 = \theta_2 = 0, \theta_3 = \theta_4 = 1$. **Then**
  - **Predict** $y = 1$ **if** $-1 + x_1^2 + x_2^2 \geq 0$
  - **Predict** $y = 0$ **if** $-1 + x_1^2 + x_2^2 < 0$



Complex nonlinear decision boundary

- **More complex nonlinear decision boundaries can be achieved by including higher order terms:**

$$h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^2 x_2^2 + \theta_6 x_1^3 x_2 + \cdots)$$



Interpreting Coefficients

Interpreting Coefficients

(1) Logistic regression equation

$$\ln\left(\frac{p}{1-p}\right) = \theta_0 + \theta_1 x_1$$

with

- $p = P(y = 1|x)$
- $\dfrac{p}{1-p}$ is the odds.

(2) Exponentiate both sides

$$\frac{p}{1-p} = e^{\theta_0 + \theta_1 x_1}$$

So the odds are:

$$\text{odds}(x) = e^{\theta_0 + \theta_1 x_1}$$

Interpreting Coefficients

(3) Compare odds at two x-values

Odds at x:

$$\text{odds}(x) = e^{\theta_0 + \theta_1 x}$$

Odds at $x + 1$:

$$\text{odds}(x+1) = e^{\theta_0 + \theta_1 (x+1)} = e^{\theta_0 + \theta_1 x + \theta_1}$$

(4) Take the ratio

$$\frac{\text{odds}(x+1)}{\text{odds}(x)} = \frac{e^{\theta_0 + \theta_1 x + \theta_1}}{e^{\theta_0 + \theta_1 x}} = e^{\theta_1}$$

(5) Final interpretation:

$$e^{\theta_1} = \text{odds ratio for a 1-unit increase in } x$$

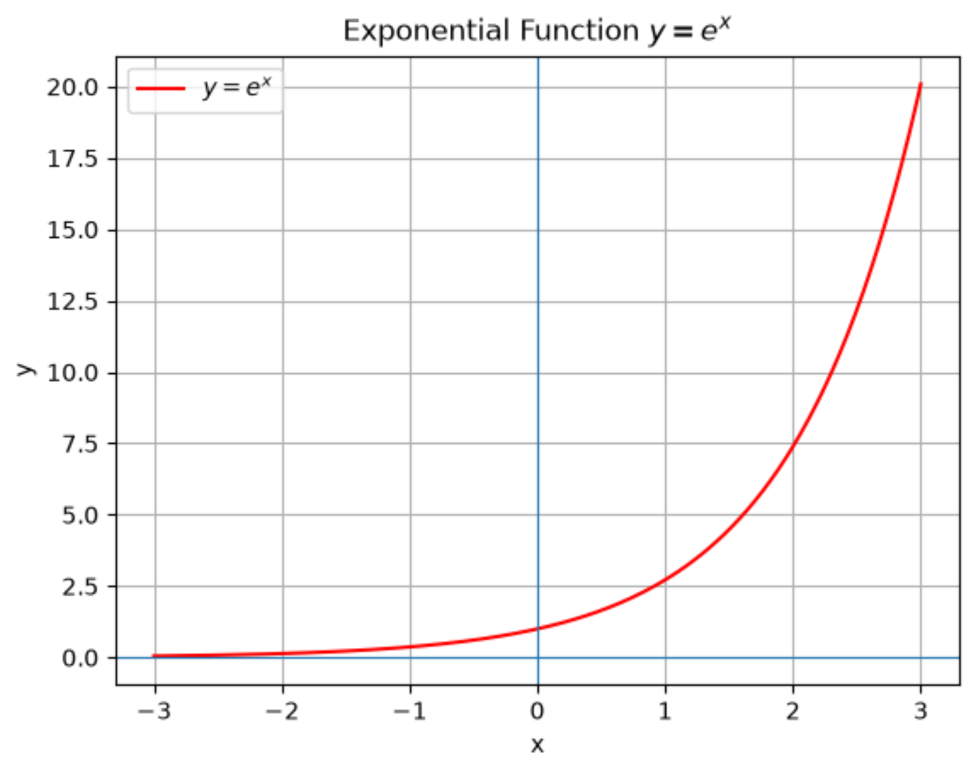
Interpreting Coefficients

Interpretation of θ_j:

- Each θ_j affects the log-odds.
- $\theta_j > 0$: increases log-odds $\rightarrow$ higher probability
- $\theta_j < 0$: decreases log-odds $\rightarrow$ lower probability

Or we can say:

- If $e^{\theta_1} = 1$: no effect
- If $e^{\theta_1} > 1$: increases odds
- If $e^{\theta_1} < 1$: decreases odds



Example

Given a 1-feature logistic regression model:

$$\ln\left(\frac{p}{1-p}\right) = \theta_0 + \theta_1 x$$

where $\theta_0 = 0$ and $\theta_1 = \ln(2)$.

So the model is:

$$\ln\left(\frac{p}{1-p}\right) = \ln(2) \cdot x$$

Interpreting Coefficients: Example

Step 1: Compute the odds when $x = 0$

- Log-odds:

$$\ln\left(\frac{p}{1-p}\right) = \ln(2) \cdot 0$$

- Odds:

$$\frac{p}{1-p} = e^0 = 1$$

- So:

$$\text{odds}(0) = 1$$

This corresponds to $p = 0.5$.

Interpreting Coefficients: Example

Step 2: Compute the odds when $x = 1$

- Log-odds:

$$\ln\left(\frac{p}{1-p}\right) = \ln(2) \cdot 1 = \ln 2$$

- Odds:

$$\frac{p}{1-p} = e^{\ln(2)} = 2$$

- So:

$$\text{odds}(1) = 2$$

This corresponds to $p = \dfrac{2}{1+2} = \dfrac{2}{3} \approx 0.667.$

Interpreting Coefficients: Example

Step 3: Compute the odds ratio

$$\text{odds ratio} = \frac{\text{odds}(1)}{\text{odds}(0)} = \frac{2}{1} = 2$$

Step 4: Compare with e^{θ_1}

$$e^{\theta_1} = e^{\ln(2)} = 2$$

Final interpretation:

When $\theta_1 = \ln(2)$, every time x increases by 1 unit, the odds ratio is 2 and the odds multiply by 2.

Interpreting Coefficients: Example

- Now let's add when $x = 2$ and $x = 3$ for the exact same model with $\theta_0 = 0$ and $\theta_1 = \ln(2)$. So
$$e^{\theta_1} = e^{\ln(2)} = 2$$
Meaning that each $+1$ increase in x multiplies the odds by 2.

- Compute odds for $x = 0, 1, 2, 3$.
$$\text{odds}(x) = e^{\ln(2)x} = 2^x$$
So:

x	log-odds	odds
0	0	$2^0 = 1$
1	$\ln(2)$	$2^1 = 2$
2	$2\ln(2)$	$2^2 = 4$
3	$3\ln(2)$	$2^3 = 8$

Interpreting Coefficients: Example

- Convert odds to probability

$$p = \frac{\text{odds}}{1 + \text{odds}}$$

x	log-odds	odds	probability p
0	0	$2^0 = 1$	$\frac{1}{2} = 0.5$
1	$\ln(2)$	$2^1 = 2$	$\frac{2}{3} \approx 0.667$
2	$2\ln(2)$	$2^2 = 4$	$\frac{4}{5} = 0.8$
3	$3\ln(2)$	$2^3 = 8$	$\frac{8}{9} \approx 0.889$

Interpreting Coefficients: Example

- Now compute the odds ratio between consecutive x values:

$$\frac{\text{odds}(x+1)}{\text{odds}(x)}$$

From $x \to x+1$	odds ratio
$0 \to 1$	$\frac{2}{1} = 2$
$1 \to 2$	$\frac{4}{2} = 2$
$2 \to 3$	$\frac{8}{4} = 2$

So the odds ratio is always 2, which equals:

$$e^{\beta_1} = e^{\ln(2)} = 2$$

Interpreting Coefficients: Example

Key points

- Odds increase by a constant factor ($\times$ 2)
  - $1 \rightarrow 2 \rightarrow 4 \rightarrow 8$
  - multiplies by 2 each step
- But probabilities do NOT increase by a constant amount.
  - $0.5 \rightarrow 0.667 \rightarrow 0.8 \rightarrow 0.889$
  - increases shrink as you approach 1.
- This is why logistic regression behaves realistically:
  - it has diminishing returns near probability 0 or 1
  - but a consistent effect in log-odds space.

Interpreting Coefficients: Example

Now let's take a look at an example where θ_1 is negative.

- We use the same logistic regression form:

$$\ln\left(\frac{p}{1-p}\right) = \theta_0 + \theta_1 x$$

  with $\theta_0 = 0$ and $\theta_1 = \ln(0.5) = -\ln(2)$. So θ_1 is negative.

- Compute the odds ratio

$$e^{\theta_1} = e^{\ln(0.5)} = 0.5$$

  meaning that each $+1$ increase in x multiplies the odds by 0.5, i.e. cuts the odds in half.

Interpreting Coefficients: Example

- Compute odds

$$\text{odds}(x) = e^{\ln(0.5)\cdot x} = (0.5)^x$$

x	log-odds	odds	probability p
0	0	$(0.5)^0 = 1$	$\frac{1}{2} = 0.5$
1	$\ln(0.5)$	$(0.5)^1 = 0.5$	$\frac{0.5}{1.5} \approx 0.333$
2	$2\ln(0.5)$	$(0.5)^2 = 0.25$	$\frac{0.25}{1.25} = 0.2$
3	$3\ln(0.5)$	$(0.5)^3 = 0.125$	$\frac{0.125}{1.125} \approx 0.111$

Interpreting Coefficients: Example

- Compute odds ratio

$$\text{odds ratio} = \frac{\text{odds}(x+1)}{\text{odds}(x)}$$

From $x \to x+1$	odds ratio
$0 \to 1$	$\frac{0.5}{1} = 0.5$
$1 \to 2$	$\frac{0.25}{0.5} = 0.5$
$2 \to 3$	$\frac{0.125}{0.25} = 0.5$

So the odds ratio is always 0.5, which equals:

$$e^{\beta_1} = e^{-\ln(2)} = 0.5$$

meaning that each 1-unit increase in x cuts the odds of class 1 in half.

How to Learn the Parameters?

Cost function

- The training set consists of m examples:

$$\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \cdots, (x^{(m)}, y^{(m)})\}$$

- Let

$$x = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix}$$

with $x_0 = 1, y \in 0, 1$.

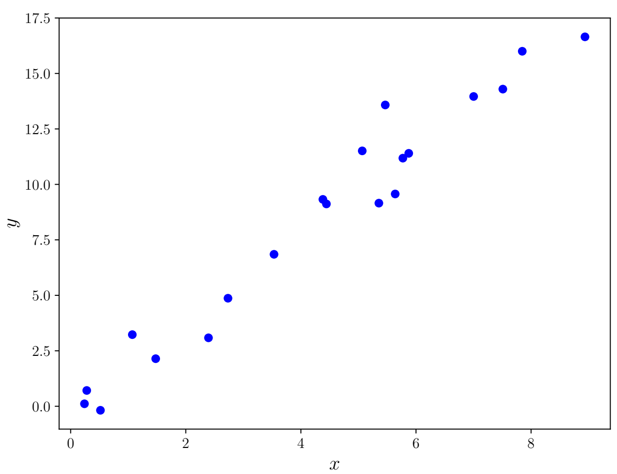
- Use the following logistic function to determine y:

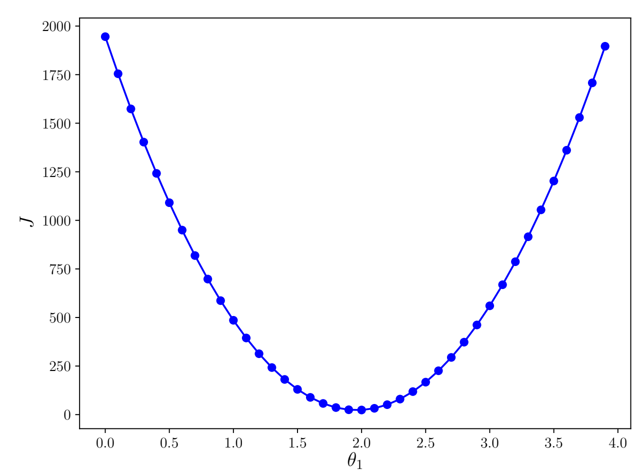
$$h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}}$$

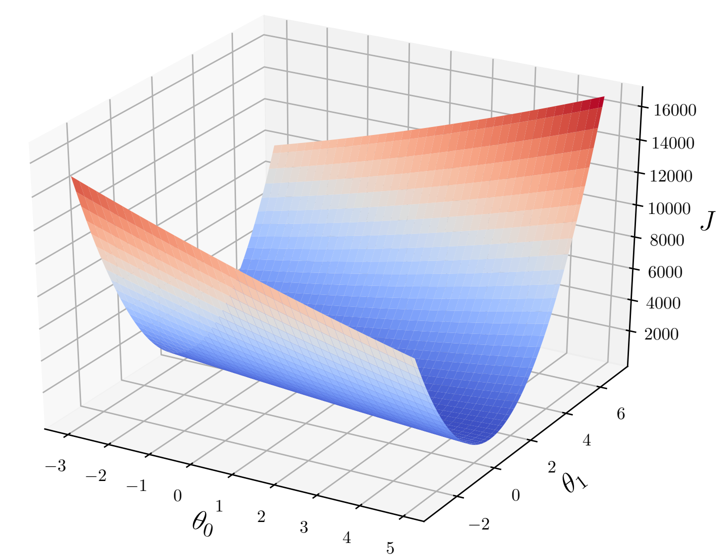
Cost function for linear regression

- Cost function for linear regression $h_\theta(x) = \theta_0 + \theta_1 x$:

$$J(\theta) = \frac{1}{m}\sum_{i=1}^{m}\frac{1}{2}(h_\theta(x^{(i)}) - y^{(i)})^2 = \frac{1}{m}\sum_{i=1}^{m}\frac{1}{2}(\theta_0 + \theta_1 x^{(i)} - y^{(i)})^2$$







- However, this type of cost function is not appropriate for logistic regression since the cost function

$$J(\theta) = \frac{1}{m}\sum_{i=1}^{m}\frac{1}{2}(g(\theta_0 + \theta_1 x^{(i)}) - y^{(i)})^2$$

with $g(\cdot)$ being the logistic function might not be convex.

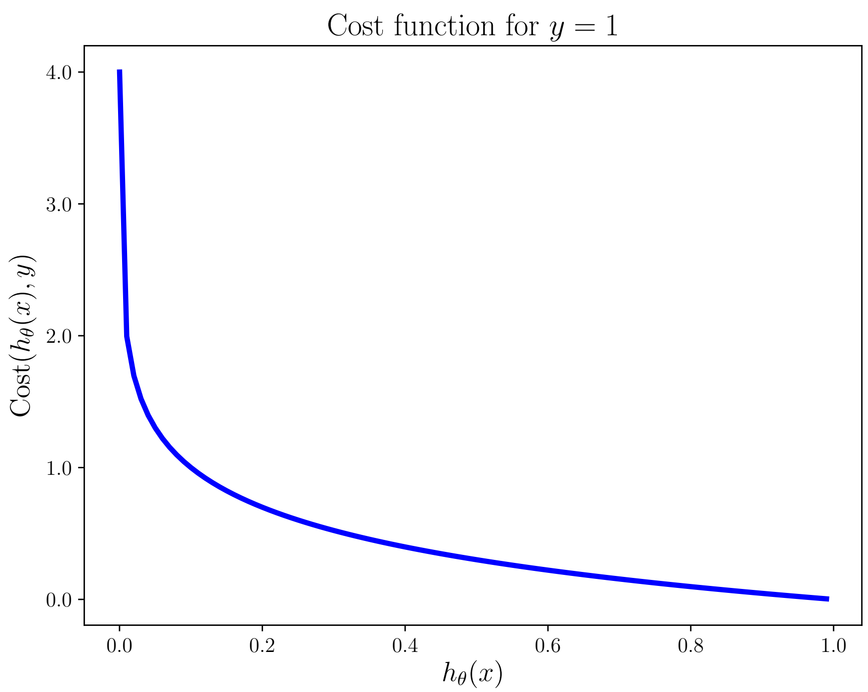
Cost function for logistic regression

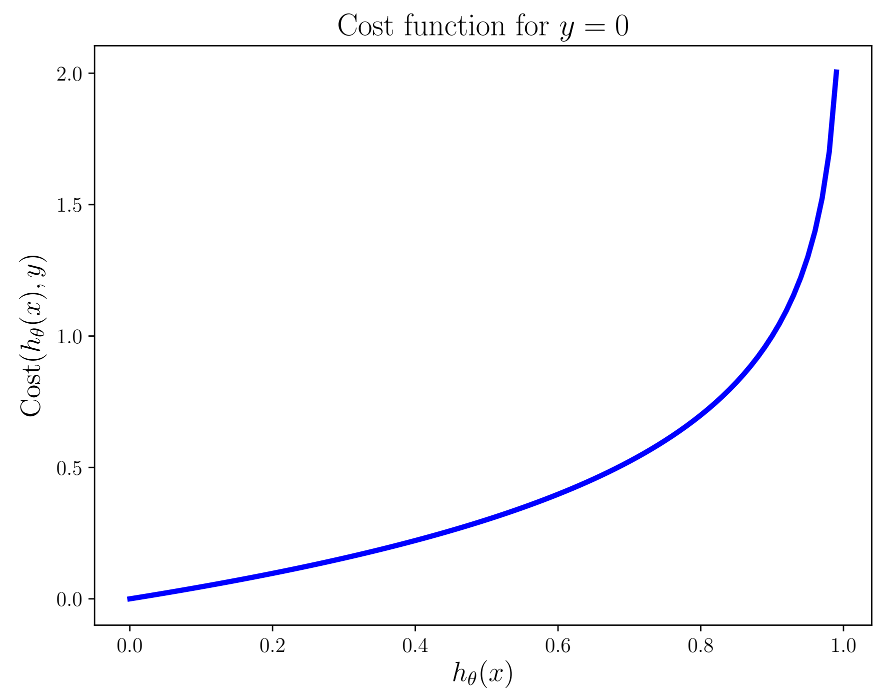
- Define

$$J(\theta) = \frac{1}{m} \sum_{i=1}^{m} \text{Cost}(h_\theta(x^{(i)}), y^{(i)})$$

with

$$\text{Cost}(h_\theta(x^{(i)}), y^{(i)}) = \begin{cases} -\log(h_\theta(x^i)) & \text{if } y = 1 \\ -\log(1 - h_\theta(x^{(i)})) & \text{if } y = 0 \end{cases}$$





Cost function for logistic regression

- The logistic cost function defined for $y = 0$ and $y = 1$ can be combined into one equation as follows:

$$\begin{aligned} J(\theta) &= \frac{1}{m}\sum_{i=1}^{m} \mathrm{Cost}(h_\theta(x^{(i)}), y^{(i)}) \\ &= -\frac{1}{m}\sum_{i=1}^{m} \left[y^{(i)} \log h_\theta(x^{(i)}) + (1 - y^{(i)}) \log(1 - h_\theta(x^{(i)})) \right] \end{aligned}$$

- To fit parameters θ, we minimize $J(\theta)$ to get

$$\mathrm{argmin}_\theta J(\theta)$$

- After training, predictions can be made for a new given x as:

$$h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}}$$

that is interpreted as $P(y = 1|x; \theta)$.

Gradient descent

- To minimize

$$J(\theta) = \frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log h_\theta(x^{(i)}) + (1-y^{(i)})\log(1-h_\theta(x^{(i)}))\right]$$

We simultaneously update

$$\theta_j := \theta_j - \alpha\frac{\partial}{\partial\theta_j}J(\theta)$$

and repeat the procedure until convergence. The partial derivative term is:

$$\frac{\partial}{\partial\theta_j}J(\theta) = \frac{1}{m}\sum_{i=1}^{m}\left(h_\theta(x^{(i)}) - y^{(i)}\right)x_j^{(i)}$$

Gradient descent for linear regression

- Linear model:

$$h_\theta(x) = \theta^T x = \theta_0 x_0 + \theta_1 x_1 + \cdots + \theta_n x_n$$

- Cost function:

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right)^2$$

- Gradient descent by simultaneously updating θ and repeat the procedure until convergence:

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta) = \theta_j - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right) x_j^{(i)}$$

Gradient descent for linear regression

- More specifically:

$$
\begin{aligned}
\theta_0 &:= \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta) = \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right) x_0^{(i)} \\
\theta_1 &:= \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta) = \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right) x_1^{(i)} \\
&\vdots \\
\theta_n &:= \theta_n - \alpha \frac{\partial}{\partial \theta_n} J(\theta) = \theta_n - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right) x_n^{(i)}
\end{aligned}
$$

Thank you!